Т4 $\xi \sim p(x)$

$p(x) = \theta\{(-1;0)\cup(0;1)\} + p_1\{0\} + p_1\{2\}$

$\int_{-\infty}^{+\infty} \overset{dF(x)}{\cancel{p(x)dx}} = 1$ (нормировка)

$2\theta + 2p_1 = 1$

$p_1 = 0{,}5 - \theta \quad (0 < \theta < \frac{1}{2})$

$p(x,\theta) = \theta\{(-1;0)\cup(0;1)\} + (0{,}5-\theta)\{0\} +$
$(0{,}5-\theta)\{2\}$.

① Метод моментов.

$\alpha_1 = M\xi = \int_{-1}^{1} \overset{xdF(x)}{\cancel{x p(x,\theta)dx}} = \int_{-1}^{1} x\theta dx + 2\cdot(0{,}5-\theta) =$
$= 1 - 2\theta$

$1 - 2\tilde{\theta} = \overline{x}$

$\tilde{\theta} = 0{,}5(1-\overline{x})$.

· $M[\tilde{\theta}] = 0{,}5\, M[1-\overline{x}] = 0{,}5 - 0{,}5(1-2\theta) = \theta$

$\tilde{\theta}$ - несмещенная

· $D\tilde{\theta}$:

$\alpha_2 = M\xi^2 = \int_{-1}^{1} x^2\theta dx + 4(0{,}5-\theta) = \frac{2}{3}\theta + 2 - 4\theta =$
$= 2 - \frac{10}{3}\theta$.

$\cancel{D[\tilde{\theta}] = \alpha_2 - \alpha_1}$ $D\xi = \alpha_2 - \alpha_1^2 = 2 - \frac{10}{3}\theta - (1-2\theta)^2 =$

$= 1 + \frac{2}{3}\theta - 4\theta^2$

$D[\tilde{\theta}] = \frac{1}{4} D[1-\bar{x}] = \frac{D\xi}{4n} = \frac{1}{4n} + \frac{\theta}{6n} - \frac{\theta^2}{n}$

$D[\tilde{\theta}] \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}$ – состоятельна.

② Метод макс. правдоподобия.

m – число раз, сколько в выборке встретилось „0" или „2".

$L(\theta) = \theta^{n-m}(0,5-\theta)^m$

$\ln L(\theta) = (n-m)\ln\theta + m\ln(0,5-\theta)$

$\frac{\partial \ln L}{\partial \theta} = \frac{n-m}{\theta} - \frac{m}{0,5-\theta} = 0$

$\frac{n-m-2\theta n}{\theta - 2\theta^2} = 0$

$2n\theta = n - m$

$\tilde{\theta} = 0,5 - 0,5\cdot\frac{m}{n} = 0,5(1-\nu); \ \nu = \frac{m}{n}$

$(\ln L)''_{\theta\theta} = \underbrace{-\frac{n-m}{\theta^2}}_{\le 0} \underbrace{-\frac{m}{(0,5-\theta)^2}}_{\le 0}$ $\quad n \ge m \Rightarrow n - m \ge 0$

$\Rightarrow (\ln L)''_{\theta\theta} < 0 \Rightarrow$ т. максимума.

$\cancel{M\nu =}$ $M m = n \cdot p = n(1-2\theta)$ $(m \sim B(n,p))$.

$p = P(\xi=0) + P(\xi=2) = 2\cdot(0{,}5-\theta) = 1-2\theta$;

$M\nu = \cancel{M m}\ M\left[\frac{m}{n}\right] = 1-2\theta$.

$M\tilde{\theta} = 0{,}5 - 0{,}5\, M\nu = 0{,}5 - 0{,}5(1-2\theta) = \theta$.

$\Rightarrow \tilde{\theta}$ - несмещенная.

• $D\tilde{\theta} = \frac{1}{4} D\nu = \frac{1}{4}\left(\frac{p(1-p)}{n}\right) = \frac{\theta(0{,}5-\theta)}{n} \xrightarrow[n\to\infty]{} 0$

$\Rightarrow \tilde{\theta}$ - состоятельна.

• Проверка эффективности.

1) $p(x,\theta)$ — непр. диф. по θ на R

2) $\int_{-\infty}^{+\infty} \cancel{\frac{\partial}{\partial\theta} p(x,\theta)dx}\ d\frac{\partial F}{\partial\theta} = \int_{-1}^{1} 1\cdot dx - 1 - 1 = 0$.

$\frac{\partial}{\partial\theta}\int_{-\infty}^{+\infty} \cancel{p(x,\theta)dx}\ dF(x,\theta) = \cancel{\frac{\partial}{\partial\theta}\left(\int_{-1}^{1} dx\right)} + \frac{\partial}{\partial\theta}(1) = 0$.

3) $I(\theta) = \int_{-\infty}^{+\infty}\left(\frac{\partial \ln(p(x,\theta))}{\partial\theta}\right)^2 \cancel{p(x,\theta)dx}\ dF(x,\theta) =$

$= \int_{-1}^{1} \frac{1}{\theta^2}\cdot\theta dx + \frac{1}{(0{,}5-\theta)^2}(0{,}5-\theta) + \frac{1}{(0{,}5-\theta)^2}\cdot(0{,}5-\theta) =$

$= \frac{2}{\theta} + \frac{2}{0{,}5-\theta} = \frac{2}{\theta(1-2\theta)}$

$I(\theta) > 0$

$I(\theta)$ – непр. на $(0; \frac{1}{2})$

$\Rightarrow$ модель регулярная.

· Метод правдоподобия:

· $\tilde{\theta}$ несмещенная

· ВМ регулярна

· $D\tilde{\theta} = \frac{\theta(0,5-\theta)}{n}$ – огр. на $\forall$ компакте из $(0; \frac{1}{2})$

$\Rightarrow D\tilde{\theta} \geq \frac{1}{nI(\theta)} = \frac{\theta(1-2\theta)}{2n} = k = D\tilde{\theta}$

$\Rightarrow \tilde{\theta}$ – эффективная.

· ОММ:

· $\tilde{\theta}$ – несмещенная

· ВМ – регулярна

· $D\tilde{\theta} = \frac{\theta}{6n} - \frac{\theta^2}{n} + \frac{1}{4n}$ – огр. на $\forall$ компакте из $(0; \frac{1}{2})$

$D\tilde{\theta} > k$

$\frac{1}{4n} + \frac{\theta}{6n} - \frac{\theta^2}{n} > \frac{\theta(1-2\theta)}{2n} = \frac{\theta}{2n} - \frac{\theta^2}{n}$

$\frac{1}{4n} + \frac{\theta}{6n} > \frac{\theta}{2n}$

$$\frac{1}{4n} > \frac{\theta}{2n} - \frac{\theta}{6n}$$

$$\frac{1}{4n} > \frac{4\theta}{12n} \quad \Big| \cdot 12n$$

$$3 > 4\theta$$

$\theta < \frac{3}{4}$ – верно; $D\hat{\theta} \neq K$, $DMM \neq DMП$

$\Rightarrow$ оценка не эффективна

(Если эффективная оценка Э, то она Э!)